Semiotics and Semantics (Discussion) ~ Lorena Sganzerla ~ Active Inference for Social Sciences 2023

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hello and welcome it is September 12th 2023 we're here in the active inference for the social sciences course at the Active inference Institute and having a discussion section on semiotics and semantics here with lenaa we're going to be having a fun discussion I hope if you're watching Ing live please feel free to write any questions in the live chat if you thought about joining and you weren't sure it's not too late otherwise we're going to pick up on some key points from the lecture and on the topic and see where we go so Lena welcome back and however you want to begin let's go from there okay hello thank you Daniel um so let's try to uh recap a little bit where we left off in during the lecture um just to remind us where we were like my point what I was trying to articulate uh was the idea of what does it mean to talk about formal semantics in a way that is not really loaded because it's an interesting thought like to have a semantic semantics that is actually only formal right in principle it would be relative to [Music] meaningful uh to meaning right in in the notion that active inference is trying to cash it out it it is it's more like an idea or a way of modeling how how we deal with meaningful situations and to understand that in a more interesting more embodied and more concrete way I was suggesting that we could use biosemiotics as uh as a as a model and as a theory for living creatures like language users to navigate the realm of symbols and gave one example which came from uh James Scott which is the European forestry that comes with a practice understanding this scientific understanding of forests as a practice that may that allows us to to refer to meaningful to meaningful things in the environment in terms of what is valuable what's interesting and what is not and what and interestingly what what is more important also from what becomes available it it is what is cropped out from the miningful perspective so things natureal uh parts of nature becomes left out or understood as something that is not desirable and you just understand what you don't want in your articulating Principle as weeds or something that you should get rid of in respect to what is more desirable in terms of crops and how you understand nature in a more commoditized way and I find like this example from Scott kind of very very uh elucidating uh in in the book he creates this in the book he's talking about States how we articulate life around one specific notion of state but in this specific example he has the idea of one official Observer and this official Observer establishes a principle of meaning in a way you have to have a handle of Nature and that that also articulates how Villages and how practices around Agriculture and and how people relate to each other within these practices gets organized specifically for uh one interesting point that he makes that is to appreciate the construction of that is like a Vision that notices not what within the meaningful crops but which is Left Outside the field of vision and I bring that specifically to understand how the principle can or could be connected to a generalized synchronization over time in relation to those symbols and those practices in terms of Niche construction and how those those how this n construction brings along with the symbolic use of uh language uh predictive value thank you Bren want to bring up any thoughts on that or any other pieces or or I'm happy to add a comment Bren okay guess not okay um a lot to say there there there was a little bit of a double play with what is cropped on one hand what is cropped is the industrial crop that is being prepared and um and then the other side of that coin is what is cropped out and what what we crop in is the part that we're paying attention to and we can't pay attention to everything so it's it's like two sides of the same coin what we pay attention to what we include in attention in our cognitive awareness leaves the shadow which is what we exclude and um that could be like the weeds or the other ecosystem services so by highlighting the meaning or one dimension of meaning or one stance on meaning that only it's like a seesaw that becomes Salient to the extent that something else must be deprioritized so that's how we accentuate these distributions of meaning and what do we get from active inference and and free energy principle beyond for example what's provided in seeing like a state I think there's a lot we could say on that but one piece is we get a process Theory a real procedural approach to how different systems come to do sense making and cognition and action so just describing the lay of the land or the history of Forestry in Europe or um any other state of affairs in the world merely describing it may not even help us determine what's going to happen a short time later let alone a long time later or what might happen in a very different ecology whereas if we understand the processes of sense making and decision making that different kinds of cognitive agents do certainly humans also though States as cognitive entities which is what seeing like a state or thinking like a state entails we get a process-based view on what otherwise might be merely descriptive and along with that process-based view we gain an increased ability to make unique explanations predictions accounts and so on yeah we can we we can also we we can also uh understand it in a different perspective in terms of realizing what becomes sensitized and what becomes desensitized along the process so you can you can possibly model that using this kind of tools and understanding how that happens over time right I don't know how how much in terms of prediction of things that were not foreseen in these descriptions we can we can achieve in fact you can achieve some some level for sure but like that system should be decide how how that delivers a very all encompassing prediction in this kind in this cases I don't believe it can right I don't think that's the purpose as well but we can at least have which is might not be as positive as we want but I think we can understand how you can gain some understanding in this process of being more sensitive or less sensitive to specific primes and how this organization organizational principles uh how powerful they become over time yeah I'll read some comments in the chat and then Lorena or Bren feel free to just give any thoughts just upcycle Club wrote hello biomimicry can enhance biosemiotics by showing us how nature solves problems and creates Beauty through signs and codes and by encouraging us to apply these principles to our own Creations yeah the idea of coding is an interesting idea but to talk about codes I don't think we can't that's my take right I don't understand like you have we haven't coded uh any type of principles within our uh cognitive ability but uh biologically speaking however you can understand these encodings in terms of becoming more or less sensitive over time to environmental cues as the example in terms of uh I is the example of Forestry right so in that specific case this what becomes more what you become more sensitive and the more you organize your life or social life is organized around specific principles then you can make sense of these relational stories uh in instead of just leveraging one for uh difficult to explain notion of encoding in terms of social cognition because it will be hard to explain how the brain will be encoding these specific symbols but they can be cashed out in the environment relationally in terms of those organizational principles and then you have this idea of biom mree like this reiterations and how that comes along with but to explain what this mimic or this meme will be then well it's not so so straightforward I think well certainly decompiling the source code of nature is somewhere between challenging and misguided or impossible but that notion of a code script for biology has animated discussions of life going back to for example Schrodinger's what is life he basically says there's two aspects of living systems there's their organizational capacity their negentropic capacity and then there's their code script they AP periodic crystallin informational repository and then another place that coding comes up a little bit less in the source code space is predictive coding and that's in a way highly compatible with active inference and gives a deflationary account on how the semiotics are encoded rather than saying well this is what this variable in the source code means or this is what this codes for or here's a code switch predictive coding is like saying we're not even going to touch the semantics of what the sign is itself we're just going to hypothesize that the way that information about the sign is transferred is through this maximum information signaling approach called predictive coding because predictive coding arose out of video compression algorithms and only at the end of the 9s was found to be increasingly relevant for neural circuitry and so on so it's another example of where like a approach ends up deflating or reframing the otherwise semantically loaded question it's like switching out the question of like where we're going to be with how we're going to get there and instead of only focusing on the meaning of the signs first off as if the meaning of the signs could be discussed outside of their implementation and then even if we could figure out what the signs were we would then just turn around and want to know how they were transmitted and so predictive coding or predictive processing in a way short circuits that or inverts it by first looking to explain how information is transmitted that could support semiotics and then in the context of biological systems we naturally have biosemiotics yeah I I would ask the question when you talk about information being transmitted right what that actually means and I think that's a very very big topic and a very large discussion in terms of um information information Theory when specifically applying that into um biological systems I think there are some interesting people working on information that makes the case that Shannon has a point and he's really interested not in the sem in the semantics or in the meaning or anything else it's kind of interest in the problem of transmission and when you talk about transmission it's worth to ask what is being transmitted exactly in the terms of of signals even if you're capturing specific patterns and you think that the pattern is the thing that will be transmitted in that sense but that will be a worth um a worth asking question to understand like the this notion of what is actually being transmit in that sense like information Pro this information theory is focusing on this notion of the problem of transmission and then all the the concepts that come along to explain that in terms of uncertainty or noise and everything that uh impact this transmissional processes and if you're saying like we're transmitting information in this sense it is breaking this semantic code apart yes it is because like the problem is not focused on that it's the focusing on the specific correlations that we have that we call it we they were mentioning as information and how these correlations are are been issue from from the cat go from what is being picked up in one side being correlated to what comes into the other side right how that how do you lose or gain more information in that but the deeper question here is uh how how can you say that you are actually gaining or losing that information because you were defining this correlation that could be transmitted or not from the g go if you don't have that you don't have anything being actually actually communicated so with that in mind you can explain these Concepts in terms of function and then there's a bunch of theories that navigate this sense of uh signaling that is specifically only functional and then predictive coding tries to T tries to push that a little further maybe but then you have to expl explain in in a deep sense how this information is being actually transmitted and processed and I don't think that is really really as obvious and as clear as we think it is specifically when talking about this biological way of nature navigating this proc this signaling we have to talk about see because the correlation if it's def AB in you like from the beginning you then only have this kind of transmission for someone specific that can be able to understand that in the other side right well certainly whether we're talking about a chemical phermone that requires some type of receptor or something more ephemeral like language that requires more of a semantic reference frame there's no receiving the message without the sender it's like they're kind of the two sides of the phone call um I'll ask a question from the chat love evolve wrote how about cultural encoding and principles in contrast to organizational so how do we think about cultural meaning and is and is that similar or different than organizational yeah that's that's a good point right I think if you take very seriously the example from the uh the forestry is a practice then then I think he has to be organizational in that sense the meaning navigates this organizational practice this organization story that that unpacks not only communication between agents but also how how the environment will be interpreted so and in this sense we will be transmitted and repeated over time because it has a history you cannot ignore like the history of this practices being pushed further and replicated through generations right so the way like the villages will be organized in terms of that specific practice for instance can be understood in a meaningful and cultural way that organizes life in virtue of some specific principles that sensitizes our agents towards specific directions and I think I don't know if I don't know if you get my point uh for instance you can talk uh if you think about archaeological findings and how uh archaeological descriptions of specific Villages and how they navigate uh this mapping out this this uh these archaeological uh sites they usually know exactly if you have some kind of History already from this this communities how this how that

City will be likely to be organized first of like where is the water and if the water is in certain directions probably uh there will be certain kinds of relationships with housing and communal housings and where they going to place their dead in relation to this kind of specific organization principle the water is here so Plantation will be not so far away from that and the city organizes like the village organizes a little bit in this sense and that builds up uh cultural practices at least archaeologically can be explained as such I I thought of something similar um organization being the the logistical or the oper ational structural aspects so those would be the roads that connect the forestry outposts or the circulatory system that connects the different organs of organization and then the culture is a cause and a consequence of organization organization is the enabling conditions for culture to arise okay so given this road Network what cultures arise how similar or different are they convergently what are their practices or their narratives that are shared amongst these different outposts um or in in the bodily example okay given the organization of the organs literally the organizing process or principle then what biosemiotics can arise with glucose and insulin and glucagon and what culturally can arise maybe even in terms of the generative model of encultured Agents as a function of organization and then what do those encultured agents do to the extent that they can modify organization and so these kinds um unrolling cause and consequence relationships can be hard to disentangle because there's not like a clean single starting point that can be separated out or because it unfolds historically it's not exactly a reaction in a test tube so how how does active inference give us an entry points into thinking about this yeah that's one one interesting thing about active inference that you you don't really need one uh you don't really needed An Origin point right you can start start modeling and start understanding that at any any given point and build up from from that story and see how it unfolds over time I think that's the most beautiful idea that you you define your parameters and then you see from that par parameters that can be arbitrarily defined and you can navate how things change within those parameters in the generated model right when you when you have your like dynamical systems for instance you have your state space they have some specific range of defining parameters that Set uh the limits of that possible states that you're going to navigate and if you use this notion that these two densities brings about in terms of biiian modeling you can use this these kinds of tools to understand how from that specific parameters update why what we narrate in active inference as if belief updating can change over time in these terms that's a very nice point it reminds me of Bruno Lor's actor Network Theory or ants which kind of Advocates a similar approach to what you describe which is like enter anywhere open-mindedly because in an iterated modeling or iterated analysis setting the first proposal or the first Trace detected we know that's not the whole picture there may not be a whole or only picture so we need to answer turn and iterate and then with active inference specifically because of the compositionality that we get within a layer Collective behavior and then also the kind of nesting smaller things inside of bigger things we can always build something and then we can go oh wow it's like there's a lot of top- down influence maybe there's a higher order factor or slower factor or a contextual Factor we should consider or oh we were abstracting across these smaller things inside but for a later day or if relevant we can dive in and so then it allows a light hold or a grasp on the proposed generative model obviously it's map not territory but even beyond that we always know that we can expand Lally and across scales starting from anywhere in an iterated modeling process without our first or even 11th model being proposed as like the accounts of some complex cultural phenomena which we're always going to only ever have our own perspectives on yeah I think you don't reallyy that in so all ENC compassing like you have one model that will model them all you can you you can navigate those models in these More instrumental terms it raises different questions when you put things like that in terms of duology of models and uh models and theories and Sciences but you they can be also described as uh tools for describe tools for yeah TOS for for grasping in describing phenomena in a more non-committal way uh how much that how much that Muggles in uh philosophical or Mor monological problems it's is still I think under like debate but it is still it's still uh it is still to to bring insights to think about specific Collective uh phenomena which is not it's it's complicated to because I think what I'm getting at I'm kind of like going back and forth but when you talk about social sciences in modeling usually you have like ways of model modeling agents or ways of modeling collectives and active influences trying to to come up with a a modeling in which you can understand agency and collectives within the same modeling uh schema and I think specific specifically for these cases if you're interested in understanding uh types of cultural phenomena among language users and uh in a specific contexts you can at least partially understand or have or gain some insight in this agent Collective relationship that that is a little bit more interesting than we have found so far in the social sciences because you can have like or or specific understandings of memes but then how the agent where is the agency in that you you lose a little bit of agency when you're trying to understand uh only agents then you don't know how that kind of scales up because active influence tries and offers this scale free uh modeling tools that facilitates this this um conversion into agents agents and uh collectives I would be interested in asking you actually how how do you not navigate that when you talk about social phenomena in ants or termites or because then yeah you have very specific coding demarcations I guess for this uh social environments yeah it's a great question it's also whether it's a uh open wounds or a a dirty closet that's just shut there's actually a lot of ambiguity and unresolved tension around well is the ant calling social are they social insects that reifies the nestmate as the individual and then what they're doing is like when we hang out and talk so reifying the nesme as individual makes the The Colony social but if the colony is individual then what nestmates are doing with each other is not social anymore than we would describe the liver and the kidney as having like a social single cellular interaction but we know that would be missing the point and so there definitely is attention with some of these evolutionary lineages like ants being monophyletically obligate usocial all and only ants are usocial they all live in the colonies there's no free living nestmate whereas other parts of the evolutionary tree there are life histories that do look a little bit more like social collaboration that can then dissolve and so knowing what level of lock in makes something social when are we socializing and what level of lockin goes beyond social and it kind of like makes when the blanket is so strong around the social that like for multicellularity that it it's kind of losing the social feature again because if we just use social to mean the connective tissue or the kind of lubricant or or the spaces between everything then it means nothing so a real tension is what space does the Social account for and is this going to be a concept with a lot of creep looking well it's a social relationship between the roots of the tree and the fungus and it's a social relationship amongst the molecules in the test tube is that too broad of a social conception are there downsides or are there trade-offs associated with a broader social concept do we dilute what the social means and then on the other side C might we exclude phenomena that could importantly be considered social well that's not social they're just talking on a train it's like well that's the other side if we were to have carving out encounters that we do want to include as social and is there a true social does active inference help us sidestep or at least negotiate this territory because we could develop a generative model or an ecosystem of shared intelligence with one person and two Internet of Things devices and a bird and a metronome and just say we made what we made and then leave it as a second order interpretation whether that is a social setting I mean I think you you pointing at like how the granularity of those scales right if you have model that one to one to everything that you already have it's probably not interesting and not informative because you cannot like Define those limits of what you're trying to understand and in the other hand when you do that when you define those limits if you're too narrow or you're going to leave out you're going to idealize also in a way right and leave out many other parts of of the system that you're interested in but if you don't do that if you don't if you don't make this this if you don't make this decision when when modeling you think then how how you can have any access to to the phenomena right and and to connect this to the forestry so in the forestry example the um deserada was Lumber now that may have been a very um narrowly scoped desire but it's it's a relevant thing to to want this resource and then industrialization science as a process professionalization those allowed the description and and the handling and ultimately the kind of top- down um legibility and control thinking like a forestry company now with respect to the social sciences by analogy to forestry science so now we're entering an era of increased legibility and control of systems and human natural capital and attention as a resource trust as a resource in a way is being cropped analogous to produce the content of lumber but now there are social content so as the social goes from being just only simply the water we swim in not something that's even compared counterfactually to Alternative cultural possibilities per se but falls under the scope of the professional management like forestry then that's when this discussion of well what's the lumber and what is the weed that are being pruned and what ecosystem Services when we pay attention to certain parts of Human Social culture and when we seek to make more of this kind of lumber what else does that come along with and and what decisions come into play and that comes along when you see process or Instagram I don't know Twitter after what happened to Twitter but um but yeah the in terms of attention uh pruning you do have that very effectively already happening you already you can see how that happens if efficiently in the voting elections for instance it can be very very easily manipulated and we saw what happened in elections for instance in Brazil or in the US and well all over uh you navigate a system of uh attention organization that selects in only what he thinks that you're interested in in the effect of that in terms of decision making and in terms of trust in terms of um diversity of discourse we have seen that it can be devastating the the and the question would be like how can you make this a little bit how can you use active inference and these tools that we have to make this kind of Technology a little bit more inclusive without being super also disruptive is it even like a possibility I I really don't know I think it's an ethical problem that we have when we talk about social systems and these principles of attention exclusion leaving things out lot to say on that um with modeling and interpolating and models whether it's large language model type or linear regression type it is able to interpolate to put data points within um an enclosure of what is already been described and basian statistics and other types of prediction algorithms are not immune to that either if we have some prior distribution belief um we believe it's possible for it to be between 20 and 30° outside then when we get data inside of our bounds we can converge and that actually tightens our beliefs not always a bad thing of course and then outside of the scope of the possible those data are just seen as noise and they may not even be integrated so then that can lead to a reinforcing process not necessarily an Adaptive One either where information that's confirmatory is used to ratchet and increase precision and then information that is not confirmatory is discarded as not useful it doesn't yeah in this case yeah it doesn't even enter the uh the landscape Road of attention of the user in what I what I'm what I would thinking in relation to this no this idea of uh of Forestry is because we have now organizational principles that are faster and more populated with uh information and attention grabbing kinds of uh prompts we transform the same um we just from the same principle into um a harder to grasp process it's it's hard it becomes more complex it becomes harder to describe this the process of understanding that in terms of Forestry is kind of way more grounded when you apply that socially to this kind of environment that we're talking like internet or this highly uh volatile um uh and virtual in principle environment it becomes much harder to assess what is being pruned out or not you just don't know it makes it more clear that there is a process of pruning you need to have no control of what is being taken out you have no idea what's being like left away from your perspective of attention so to this topic of intended or unintended semantic pruning I think um this has helped me identify attention if we want to talk about the semantics or semitics biotics whatever we want to call it of the letter E in relationship to letter A we're looking for an account that might convey letter but were generalized above enness or anus so in some ways we've erased or abstracted generalized away from the particulars of the sign's meaning by taking a Semitic perspective and so it's like we got into the game to have semantic account of biological phenomena or of any phenomena for biological entity yet this process in a sense distances us from the particulars that actually do Grant it that meaning in the actuality like if we walk into a Sacred Space and it's like oh well that is just doing that over there and this architecture is doing this and because of the layout people must use the room this way that's removed from the actual meaning which is the cause and consequence with the organization and culture that we discussed but that Prima fascia accounts doesn't generalize get any generalization that we make so that we can talk about multiple places or multiple letters on a common accounting framework is going to remove those particularities so it's a very challenging Catch 22 whether we want to really account for the N equals 1 and maybe even live and act and play and work in the N equals one versus develop something that might be satisfying and cut and dry but it's always going to be on the sidelines with respect to what it actually means but what do you mean like is I don't understand what you what you're getting at like is it you mean like is generalized or not generalized it I didn't get that like when we have a general when we generaliz when when we ask what words mean what what words mean Beyond any given word we've lost touch with what any given word means but words are what they actually mean in a situation so it's just in this scientific approach to meaning semiotics and semantics we have these two tensions or a tension to reconcile between actually giving an account of the meaningfulness of the specific thing versus taking a more scientific approach which would be gathering a group of similar things and finding their shared attributes which out of necessity must abstract must crop out features that are are empirically or um anecdotally taken to not matter like we abstract whether it was a Monday or Tuesday that we did experiments we just combine all of our replicates together when we do the scientific analysis you you mean you mean if you generalize the practice you lose what you being if you generalize it too much you lose everything that's meaningful to you in terms of like for you formalize it and then you have yeah you have an empty semantics you have a semantics with no meaning that can only be meaningful in terms of how you apply that in terms of specific practices yeah it's like I I have a framework for you know the formal meaning of letters within words within sentences within par graphs and then it's like well but now what's your next n equals one word that you're actually going to bring it's not just about building some abstract scaffold yeah for bigger and bigger Boulders of meaning at some point through our embodied thought and action in our experience it is going to come down into the particulars and it it doesn't immediately seem clear that simply generalizing further brings us there yeah yeah I particular don't I don't really believe you can have that kind of generalization I think we attempt to build whatever what I see that's the interesting aspect of the formal thematics you try to build this more General uh understanding that's just formalizes to model but for that to be meaningful it has to be embedded in a way of practicing it like in a forestry practice you have to be embedded in a cultural environment in that can be actually uh observed in terms of what happens when a generative model is being updated or not is that what you mean because otherwise I don't really understand how it becomes too yeah it becomes too formal it becomes too abstract in terms of some I would not understand what that is I I'll think of a little more um Bren or TJ do want to add any thoughts questions yeah sorry I I think like I don't have a lot of context I've been like following some of the discussion but not attended a lot of these calls I'm like it's all good yeah um yeah I don't know uh I think um yeah

I'm in general just sort of interested in trying to understand the relationship between um yeah like sorts of like constructionist epistemologies and um yeah like generative models and active inference um style stuff and um how active inference style like yeah what are the philosophical commitments that active inference involves with respect to knowledge and epistemology and there's a sense in which like rational Choice models often assume a certain sort of like um objectivity to beliefs and then desires are where most the degrees of freedom lie or something um whereas there's a lot of like issue like there's a lot of things that come up once you start talking about reflexivities where Concepts themselves become self-fulfilling prophecies uh what you believe about what your beliefs about society and sub shaping the society itself so there's a sense in which there's this codependence between the object of knowledge and knowledge about that object itself and I I think yeah like I'm sort of interested Ed in like how the active entrance ways of thinking about this offers what perspectives on those questions and stuff like that um yeah thank you great questions yeah so when you talk about the uh object of knowledge and the knowledge itself and active inference uh I think you you you can think of in terms of this well there's a few way to think about that I guess uh the knowledge itself it has to come epistemically from where your agent is actually in invaded and situated specifically in active it's embodied he kind of needs to have a model the model is modeling a body is modeling a body that is in the world and it's all the time uh situated is never disconnected to its environment and it's his and how the history of that environment right yeah uh but if you think if you push that to the object of that knowledge uh I don't know how how different um that will how different how can you different differentiate that so sharply when you talk about active imprints because if you think of uh in terms of a generative model did what will be updated comes is updated from a prior that is always directed in this body in this situated uh realm so they sort of conflate as I see maybe maybe mistaken but they sort of conflct in how you going to update your generative model which it will be at the same in at the same time what you know and what you they object of this knowledge so I I sorry if I can just like uh raise a question to that yeah so I think like I think that there is a sense in which like with respect to something that in epistemology would be called self- knowledge there is there is a question here that the the active inference agent has this model of that is like situated in the body or like in in its like um like structure and is also modeling that structure itself and there is like this issue of like where where yeah like the belief is not unidirectionally dependent on the object sort of a thing but has this feedback loop thing comes up there but there's another way in which I think like uh where this sort of thing comes where even if the object is outside of the body of the knowledge I think that that that could still come up so for instance if I live in a if if an active influence agent lives in a society of other active influence AG where currently all of them defect on prison's dilemma but maybe like on further reflection they would want to like again like I don't know what further reflection would mean in the active inference sense but I'm like just saying in the intuitive sense like that what would it mean for them to like go from one fix point to another fixed point where they're all cooperating or or or something like that and that that I think is not clear to me like how that shift can happen in active inference where it's not just purely using the empirical information to keep running the the the mill of the if of your like free energy equation but also recognizing the contingencies and then identif in those contingencies to make something like shifts from one state to another like if if I model that the society is a cruel space and if everyone models the society is a cruel space then we will end up creating a society that is a cruel space but if only we could recognize that our belief or our model that Society is a Cru space is a contingent belief and we could all somehow negotiate or communicate and shift to a different belief about what Society is and Society is a kind place then we will end up creating that kind of society or something and that sort of like a belief about the thing that Society is it's not entirely within the agent like it's sort of it it it has a weird inter subjective character so it's not entirely outside of the agent either but it's not entirely in like internal to the agent either or something yeah you can think of that like you're talking about active inference action comes first right you have to you don't have the knowledge like inside your head waiting for you to to push that out of the action has to come first so if you if you take that as a principle an organizing principle it depends on the contingency of your actions the possibilities might be limited by your situated um condition but they are limited it doesn't mean that you don't have like a range of options from that state space in which you can go you can navigate so you can navigate it for the good or for the bad but it starts with the the the action perception Loop starts with the action all out a few other pieces on that TJ very good point like when we see the pragmatic component of action as self-fulfilling epistemic component of action having to do with gaining knowledge and the salience of information but the pragmatic component is about bringing our observations into alignment with our preferences and then you identified two related settings one of them is like pure intra subjectivity like I expect myself or I find myself or I habitually think about X well that's kind of a trivial self-fulfillment because thinking about something is its own fulfillment and then when we're in this second person or social space we can agree on little performance art or on a game and games give evidence to that that when people are able to perform communication or message passing such that that new expectations or preferences are set whether more top down like more seeing like a state or more laterally more like Collective Behavior then those regimes can be embodied and then you brought up a very important point about the awareness of the contingencies or aware awarenesses of adjacent posses or of counterfactuals so the body might be um we could model its more morphology as doing some computation but that doesn't mean that the the morphology has an awareness or can explore a counterfactual like bodies are not always exploring every counterfactual um because skin is in the game but again mentally it's very cheap and easy to consider counterfactuals and situational awareness of social settings especially if it can be not understood but communicated opens up a space where we're all playing prisoners dilemma now there's parameterizations of prisoners multi-agent prisoners dilemma where tragically we just all defect all day and never learn there's also parameterizations where you just start all cooperating and never defect and then the interesting settings are where there might be like phase transitions or where a perturbation moves the system from one attractor into another attractor so there's just a lot of richness and that's the work of building generative models in that iterated modeling process to to unpack and refin and active inference isn't going to have a stance on the prisoners dilemma we can use active inference to to describe the prisoners dilemma but active inference isn't going to come down on on literally even a little 2 by two Game Theory Matrix so we can then extrapolate that active inference is not going to come down with an opinion on something significantly more rich and real than a 2 by two Game Theory Matrix and so that work is what people enact in applying active right and I think like my question is something like but in so far as like humans try to constantly improve their games how can we model that sort of a prior over your actions or policies uh that enables that sort of like Progressive um um Improvement or something well you you can we we're assuming that we're trying to improve right we we we don't know that but we can model them as improving but like following Daniel's lead you we're talking about these specific games and contingencies they they they this these transitions from one state to another specifically if you're talking about this modeling of social actors and talking about several actors if your model involves several of them will not be really linear right you don't have like a easy smooth transition of improvement you can capture over time modeling how they might tend to go towards a direction and understand that as Improvement but that takes specific iterations that will not be fully linear I guess specifically if you want to understand error and Improvement instead instead of try and error but tryer and adding up into this eror like error

Improvement but when you're modeling specifically with dynamical systems for instance you you you're not going to see a full linear story and I don't think you're going to see that uh if you're modeling using active inference and multiple agents agents I'll add one piece there so H how can we go about modeling multiscale social goals so we could think about you know different citizens in a region and they the model is going to choose what knowledge to realize in the generative model like what you put into the calculator it's going to calculate upon now this happens to be a elaborate stochastic method but if we only tell people yeah there's just your personal income optimize for that your preference is to have that be higher well then the simulation is going to unfold a certain way if we say well there's your personal financial income and then there's your city and then there's your region then then that opens up through through different um mechanisms you know maybe an individual goes flat or neutral on the personal in service of a larger win at a at a higher level well now what if the modeler says it's not just about the financial income now we're also going to have this secondary Criterion well now different agents could attend differently to those two different out outcome variables and maybe there's one simulation where 50% only care about this and 50% care about that and it works perfect or doesn't work it's like a broad surveying you know or Lego set or or grammar or or active inference ontology for scoping very broadly and very inclusively but all the work is in the real simulation development and sweeping because very little could be said about a game theory game if you didn't know what the parameters were in The Matrix you wouldn't know whether this P prisoners dilemma was going to be like an always cooperate or which agents with which strategies would or wouldn't cooperate if you don't know what the Matrix looks like so before the values have kind of been filled in it's just like actually describing the adjacencies and the counterfactuals and the contingencies of our scientific model but that's still prior to having the model in hand and the model you know so now we're looking at that tree in the European forest and like well if I had this tool then I could cut it down this way if I had this tool I could cut it down this way or I could wait it's like that's alternative actions about that tree now in the social science setting discussion of different scientific methodologies are like counterfactuals about that social tree yeah and specifically if you're navigating this kind of game theoretical models if that's what you're interested in you can even Define in terms of um prisoners dilemma for instance uh specific values just see what happens when everybody def um deflects and in how frequent and modulate those frequencies in terms of specific uh in multiple iterations how how how you going to uh build up the wage or the tendency of deflecting again if your neighbor deflected once and deflected twice or three times or uh and see how that we accommodate over time how many times this agent will change its behavior until it stops changing the behavior but that you don't really but that's that that again you use you might use a ban understanding Aion model for that but it not necessarily explain um it does not necessarily explain the action fully I mean socially you just you're modeling only the interest of that specific actor I I I have a slightly meta question if I can ask both of you uh I'm not sure how these calls usually go uh I'm curious what are your interests in active inflence what do you think like what brings you to active influence and like what do you think what do you think active adds to our theoretical understanding and in what ways or something right I go for it well I think I was trying to touch upon that before and how we can if if if you're navigating for instance behavior in terms of Game Theory this kind of modeling if you're talking about like trust games for instance you're going to understand how this to addic interactions evolve over time specifically over interactions but you don't have an account of change because you have very limited options of actions and you don't have an account also own scales and I think active inference tries to add a little bit in account of how you can scale that and how uh how can I more parameters into the game and see how they evolve with a a clear um understanding some a richer understanding that for instance when you talk about specific interactions in Game Theory so when you that's one one aspect that active influence kind of contributes another point is because we have this parameterization that is specifically in terms of belief updating and also in terms of Dynamics we are always talking about uh agents that will be situated and embodied and have a very specific uh synchronization with the environment all the time because when you talk about active inference the action comes always first so you have a less you try to have a less reductive model spec and when you want to model uh cultural agents and Collective action I think that's kind of desirable that's what you're looking for is it also your area of Interest sorry is it also your area of Interest mine or is that Lorena's area of Interest Lorena's what do you mean oh sorry like my question was also like what brings you to active inference and I think you answered a lot of ways in which active inference adds theoretical value but I'm I wasn't sure like if that was also your subjective interest or motivation it's my subjective interest yeah I'm I'm curious to see how that pans out right I I I'm interested in how that helps us to to think of this type of uh phenomena and I'm I'm very interested in how this relationship from Agents to collectives and Back Again work like and I think it might give us some well you give us some very insightful way of modeling these relationships and navigating is as you want because this this uh feature of being scree is kind of helpful in to looking things you can you can Define your uh Mark of blanket in terms of are you interested right and see what happens I don't know yeah I'll just give a few thoughts even though also surely some of these thoughts are are scattered um you asked TJ what is active inference add to our theoretical account of the social it does many things in in my view and that's what makes it so valuable epistemically and pragmatically it gives us the expressivity within a scale to describe Collective behavior and across scales to describe nested systems so that brings continuity to discussions of the social with broader discussions of the the biological ecological computational physical and so on so that's a multiscale transdisciplinary element active inference provides us with legible first principles which can either be taken instrumentally as just teristics and also in certain situations can be viewed empirically as justifiable first principles and then from those legible first principles which are not English specific they're already in a space that has a lot more possibility and promise than for example a mere enumeration of important adjectives or nouns we can develop lists of important adjectives and nouns when we have a broader cognitive framework and so this methodological turn then supports just a tremendous diversity of thought and work instead of asking us to kind of go through the eye of the needle and you know get on board with this one ranking of features and then try to apply it more broadly that's like an overfit before the cart situation with actm we can expand the methodological discussion connect it to theoretical physics and Mathematics in ways that really haven't been seen before and then that opens up more than it closes down it brings up more questions than it resolves in and of itself but it opens the questions that can be resolved like what generative model can I make that will keep the temperature in this building livable but active inference isn't going to have a stance on what you should do for that building so we need to in a way take a step in the methodological Direction which I see as moving towards active inference to enable a million steps in different directions I have a bunch of questions here uh I think one is probably just a reference question uh there are three topics that I'll say if you guys have any papers or articles on those things I would love to like get a reference one is in what ways can we model a collection or an aggregation of active inference agents as a like super agent or like super like active inference super agent or something and like how how does this like decomposition and aggregation uh like flexibility work in different context like if there's like work that tries to either make informal commentary or formal modeling either ways I would love to see secondly on the first just on the first check out the previous section of this course on Collective Behavior where I reviewed many such works great thanks okay it is it on the same notion is it linked from the same notion thing all all in the same places and in the same playlist with this course same syllabus yes got it okay um the second is is there an active Insurance account of say markets or economic interaction wherein like your your prices act as some sorts of like information signals and like like can can use active inference to then talk about aggregation in of market and enties or something I'm guessing this would also be covered in the same list indeed people have explored cognitive economics and people are working on it but that doesn't mean it's a done deal theoretically or especially in practice so keep going quite it's quite experimental still but there are a bunch of people working on that I guess I I don't remember any specific paper right now but I can I can look it up in uh for to you but yeah yeah it would be great uh I I can drop my email address here yeah sure uh the third is probably more trickier than both of these which is is there an active inance account of natural language um yes thankfully colleagues like Elliot Murphy and others have explored specifically the linguistic aspects of natural spech Beach I think Lorena's lecture and area of focus generalizes a bit beyond specific natural language because we're talking about semitics and semantics in more Intermodal sense however yes there's been limited work from uh computational Linguistics and neurobiology perspective on speech okay so then uh okay so you said something like first principles stuff when you're making a commentary on what act of inance brings I have still I have I'm still often like struggling what is active inance offering and I like try to encounter different things I think I enjoy a lot of philosophical questions that the the conversation opens up but like I'm still like very ambivalent about the theory um and when I read like this new book uh which friston has not written uh but like is Anor I don't remember was the name of the book um uh I don't yeah I don't exactly remember there's a book right which is like supposedly like supposed to be the Canon and like friston is a co-author but in the preface he says so that's the 2022 par friston pulo textbook right par yeah yeah I wouldn't quite say it's the Canon since it's a short book but certainly it is a landmark moment for fields right and when I read that the one thing that bothered me was it brings up the helmos free enery thing but it's sort of argues it by Fiat like okay this is a simple principle it can Model A lot of things but that doesn't tell me why I should why I should trust it like what what is the reason to act why why does this principle lead to cognition like what like sure youve posited a very simple principle and then said hey look this simple principle can explain so many things but well it can it also explain things that don't look like cognition and like I like I'm not sure like there's an I I got an account of why that principle is like uh works or something right go for it I'll think of some other ways to say it yeah you you you can think of like modeling by construction and modeling by first principles and if modeling by first principles you have you have some limitations in which you cannot fully explain some things but the virtue of the model model and how the first principle organizes everything that comes along with the model usually might be sufficient or might be a useful a useful tool to help you explain that phenomena that based on the first principle that you were bringing along and that's one way of one part of answering this question I think it's might be a big question for the for the F yeah here here's some short little threads on that yeah we can describe maladaptive cognitive systems or adaptive cognitive systems whether something's adaptive or not is always in relationship to its Niche like the foraging algorithm that works in the desert doesn't necessarily work in the rainforest so these systems are not like intrinsically adaptive or not we want to have natural language that allows sentences that are valid and invalid and have different kinds of semantics so again this is just an expressivity framework that enables people to build cognitive models of high reliability or important settings or you could make something that looks just like a beautiful circuit board and not even worry about what the outcome is so so yeah for sure you can describe all kinds of systems that would be like um having a uh less than functional methodological system if we said well we're doing a linear regression framework but we only want positive regressions it's like well no at the methodological level we want it to be able to do any slope so here in the cognitive modeling setting we want more expressivity so that we can actually construct the ones that we're interested in and then how does this really do cognition how is it that a first principles like surprise minimization or the bounding of surprise through free energy how can that really cover so many cognitive phenomena that's the question and the play but how is it that Newton's Laws of Motion can describe so many objects being dropped off a tower how is it that basian mechanics can describe so many different priors updated and so by understanding cognitive paths as Paths of least action which is the free energy principle a wide variety of models become of a of a similar kind now become available yeah whether that's useful to a given person in a given setting like if you make an active inference model of your business it doesn't mean you're going to succeed if you make a self-driving car with active inference it doesn't mean it's going to going to work this isn't a guarantee of success but it's a method that can be utilized and we know that the method reaches different kinds of modeling outcomes and supports subsequent perspectives and models from where it reaches but we can't a priori limit where the method reaches to like only the successful ones or only biological systems then we wouldn't even have the expressivity to talk about cognitive ecosystems with synthetic intelligences um and when you're talking about cognition also we can use these kind of first principles to model aspects of cognition but you might not necessarily think that cognition is encompassing like that everywhere all the same but you can use that to model specific things that you're interested in in cognitive behavior like adaptivity does it give me any insight to understand what is adaptive adaptive behavior for instance and which kind of insights do I get are they interesting are they useful so that's what the Ava availability that this kind of model bring along so you can you don't have to believe in in the whole story but it gives you a lot of Leverage to ask ask questions and what kind of questions you'll be asking h okay so let's say that like free like active inflence and free energy allows me to talk about systems uh but does it give also give me vocabulary to then classify those systems in terms of let's say adaptivity would be one property but like there could be other properties in terms of like how much how rich the world model is or stuff like that or yeah um if you have a portfolio of generative models you could summarize them according to the TJ statistic you could summarize them according to the number of nodes you could put them into a common environment or different environments and summarize their activity in any number of ways but we can't talk about a ranking or a summarization of a portfolio of generative models that's not in hand this is a framework that lets us build those diverse generative models and then in a very open-ended way do model comparison model selection model adequacy all the kinds of complex engineering methods that we want and also I would ask you where do you see a peer or a comparison first principle you know active inference has legible first principles which are reflected in the active inference ontology and expressions using it which can be represented in natural language so that's the talking about it but it's not only represented in natural language other Frameworks have other first principles that may or may not be legible and people may choose different paths to go down for whichever number of reasons well right models like they have the virtue can compare them and see which one gives you the best Insight that you're looking for well so yeah I think like something that I could compare to active inference is like say the ban utility maximization thing and which has a different onology I'm not a fan of that either like that also I'll just that basian utility maximization is a special case of free energy where there's no epistemic value all you have is pragmatic value and so a special diminished necessarily not necessarily you have value of information and you have explore exploit tradeoffs in ban utility maximization as well like you like like if you load into your utility function if you can beg the question and load epistem into pragma you're right then all you need is pragma but in expected free energy we have epistemic and pragmatic value which is what enables us this expressivity but I think we're aligned on that right I I think like I'm somewhat like also slightly confused because in the basan utility like between the languages comparing the languages as such like in the beijan utility maximization the exploration parts that are are prioritize based on your X an um estimation of on the value of information uh whereas I'm not sure that all holds true for the act of inflence model as well that the ep the epistemic exploration is prioritized or valued in terms of some of the same uh pragmatic thing as well you yeah you you'll have a a a lot to explore and learn you can make a mod that is 8020 or 2080 or 100 Z between epistemic and pragmatic there's nothing we can say across generative models about whether active inference prioritizes epistem or not because you can make a model that's 100% or zero so sorry I I meant more like does active entrance allow us to talk about systems that the beijan utility maximization doesn't allow us to talk about I'm talking particularly about expressive power of the language yeah um like is it like are there things that the bity maximization can talk about that active inference cannot other things that act to influence talk about that b maximization cannot two short answers one short answer utility maximization must cram epistemic value into pragmatic value we gain more expressivity when we can articulate out pragmatic value which is the alignment of preferences with observations from epistemic value which which is Information Gain so that's a huge articulation and the second thing is when you're in a anything maximization framework when you're in a utility function now again begging the question how you constructed this function that you're trying to maximize but when you're in a maximization framework we have a well-known set of approaches for finding local and Global Maxima using optimization or maximization Frameworks what active inference uniquely provides is it describes the path that that cognitive thing takes as a path of least action that minimizes surprise so instead of climbing to the top of the hill oh are we on a Foothill or are we on the biggest hill we're like a ball rolling to the bottom of the hill finding the least surprising path now that least surprising path can still include novelty and Information Gain but it turns out to be a lot more tractable and connected with statistical physics and quantum mechanics when we talk about the path of least action ball rolling down the hill rather than this sort of question begag on top of a mountain climbing adventure basian utility approach but it's up to each person to feel how they want to feel yeah and I think that a little a little bit what's in the book as well when you take the low and high road for free energy principle right because how much you want to have your utility maximization function being uh disconnected to the pragma to the to the action because when you're modeling Behavior it will be hard to do that independently like you have a lot of you have way more assumptions you committing to much to a greater level of assumptions for your utility function if you don't do that because with the pragma with with the action first that pragmatize you you you have reasons to describe have reasons to believe why that information gain works the way that does you can leverage some explanations on that with without it you you have you have you you have a model that has more assumptions and I think if you have a model with less assumptions is always healthy here let's say so yeah so I think like that that that does talk about the compare compartive analysis of the expressivity but I I would also say as a meta point that like I'm not happy with either of those languages I'm like I I'm suspect that both of those languages claim certain ontological assumptions and I'm like always suspicious wait but like where are these ontological assumptions coming from a few other points we tap into an absolutely meaningful Nexus of statistical methods and message passing schemes with active inference if you can State something in the active inference ontology and construct a generative model then message passing algorithms can be deployed that cont tractably compute that model so that's absolutely non-trivial otherwise it's possible just to get wrapped up in some analytical formalism but then once you want to do that on a 4K video now you're on Square zero with the computer engineering this is not that way and then I will respond to your metap point with a metap point which is when we've enumerated the Alternatives the one that we like best personally I feel happy with I'm also open to New Alternatives arising but being unhappy when all things considered is in play why what bothers you specifically on the of uh the active yeah like I I I think like I'm still looking for I I don't know what exactly I'm looking for but I feel dissatisfied with the account given for why Helm's free energy uh least action thing like is the central thing here or something like why does it work like why does this principle yeah there's a lot of funy answers I could give there yeah that that Delta discrepancy that's curiosity and that's the space of the open and that's learning and that's your unique contribution to make and that's everything so that's not just a thorn in your shoe that's like our experience of action amidst uncertainty which is the water we're in and so whether we take a higher order narrative perspective on that discrepancy being dissatisfying or that discrepancy being satisfying I think it's pretty cool I just need to minimize my surprise there yeah and you can choose to minimize your surprise through your attention you can choose to minimize surprise at a higher narrative order like I'm satisfied and complacent that I don't understand number Theory but I use numbers to count objects around me so that discrepancy and understanding knowing that there's a thousand years of research that could be done on numbers it doesn't unsettle me but for someone else that might be unsettling and that might lead them to ruminate it also might lead to do a PhD and number Theory so just different paths for different individuals and contexts but there's always so many Technical and meta levels with ACM huh okay so I don't want to like keep you guys over time uh but I would ask one last reference based thing which is um are there active inference accounts of see like a state phenomena uh in any literature I think I heard you mention it once so I was like yeah like of this yeah check out live stream 33 series but also this course that we're literally actually enacting in with AVL is where these accounts come to roost and where we're creating a space to develop it and it's not a cathedral for us just to spectate at today but if you want to make an active inference generative model of whatever social setting when you have the model in hand a lot of your questions will be sidestepped okay where is the 33 thing that you mentioned if you go to the page with all of our Institute live streams and look in the live stream series like the paper discussions in the 33 series or a zero video with background and context and then we have a one and A2 conversation with the authors but AVL who's also the course coordinator for this course and chyos research like ail's and and colleagues work has heavily built on the seeing like a state thinking like a state optimal grasp all these different topics yeah I'm yeah I'm on your YouTube channel I'm slightly finding it hard to find the thing um if you look up live stream 033.0 or if you go to in the video description of any video where it says all live streams here go there and search for but you you'll find it and you can always email us or get in contact but yeah and I think of V paper like thinking like a state also can help you a lot to understand how how um active INF modeling is useful for the social sciences in comparison to other types of modeling he makes an comprehensive description interesting great thanks uh yeah and this is this is our first iteration on this course we are going to continue bringing new information to the table and supporting individuals who want to build generative models you know around this area whether part of an internship program or part of a course credit or different kinds of programs that we'll be able to develop but this is the real thing and we're able to work with people who want to think and do makes sense indeed any closing thoughts on TJ then I will go and then Lena with the last word uh no that this was great thanks thanks for the conversation thank you well thank you for Brandon TJ for joining Lena for the lecture this was a fun discussion we never really know where or how it's going to go but I'm glad that we could cover so many of these topics and probably raise more questions and footholds than door slams so thank you againa yeah thank you for inviting me and I think active inference like holds a lot of space for experimenting as well some things you try to see if they make sense and if they're going to hold up and open new questions and that might be an interesting aspect of uh of the framework itself thank you everybody thank you everybody for the great live chat comments there's a lot there I couldn't read it all so till next time right